

PROJECT TOPIC: STARTING AN ECO-CLUB IN JUNIOR COLLEGE

Topic Description: Establishment of a Campus Eco-Club and Execution of Two Concrete Environmental Action Projects: Zero-Single-Use-Plastic Audit and Native Herbal Garden

Educational institutions serve as prime incubators for cultivating environmental ethics, ecological literacy, and civic responsibility among youth. While classroom lectures provide theoretical environmental science concepts, actual behavioral transformation occurs when students actively participate in on-ground, measurable conservation activities on their own campuses.

This project documents the formal establishment, organizational constitution, and operational execution of an institutional **Nature Club / Eco-Club** in our Junior College, established in accordance with the National Green Corps (NGC) guidelines under the Ministry of Environment, Forest and Climate Change (MoEFCC).

The project specifically details the execution of two concrete, measurable environmental action projects carried out over a structured three-month monitoring period: **Activity 1: Campus Cleanliness & Zero-Single-Use-Plastic Audit**, and **Activity 2: Establishment of a Native Medicinal & Butterfly Garden**. The initiative demonstrates how youth-led micro-initiatives foster measurable campus sustainability and community environmental awareness.

2. Importance of the topic (विषयाचे महत्त्व)

Institutionalizing an active Eco-Club on campus provides profound pedagogical, ecological, and institutional benefits:

Key Dimensions of Importance :-

- **Hands-On Experiential Learning:** Bridges the disconnect between theoretical environmental textbook chapters and practical ecological reality, allowing students to conduct real-world waste audits and soil preparation.
- **Measurable Waste Diversion:** Converting a college campus into a segregated waste zone diverts hundreds of kilograms of dry plastics and wet canteen scraps away from overflowing municipal landfills into scientific recycling and composting streams.
- **Campus Urban Biodiversity Islands:** Creating dedicated botanical and herbal pockets provides vital nectar corridors for urban butterflies, bees, and pollinating insects amidst concrete city landscapes.
- **Cultivating Lifelong Eco-Conscious Citizenship:** Students involved in active club leadership internalize sustainable lifestyle choices—such as carrying cloth bags and stainless-steel water bottles—that persist throughout adulthood.

The club proves that educational institutions can lead by example as zero-waste, eco-friendly models for society.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The Eco-Club initiative was executed under structured organizational and environmental objectives:

Structured Project Objectives :-

- **Club Formalization:** To establish an elected student executive committee (President, Secretary, Activity Leads) under the mentorship of the EVS faculty.
- **Plastic Waste Audit Execution:** To conduct a rigorous 6-week baseline audit of plastic waste generated across the college canteen, staff rooms, and classrooms, implementing a 3-stream segregation system.
- **Medicinal Garden Establishment:** To transform a 500-sq-ft neglected campus corner into an organic botanical garden featuring 25 species of indigenous medicinal and butterfly-host plants.
- **Decentralized Composting Implementation:** To install a natural leaf and organic canteen waste compost pit, converting organic campus scraps into fertile black humus for garden fertilization.
- **Quantitative Metric Evaluation:** To record empirical data on the mass of plastic recycled, volume of compost produced, and plant sapling survival rates over a 90-day period.

The scope encompasses student volunteer mobilization, photographic logging, and quantitative weekly audit records.

4. Project Work Methodology (प्रकल्प कार्यपद्धति)

An action-research and participatory research methodology was adopted to establish the Eco-Club and execute both core projects:

Methodological Phases Followed :-

- **Phase 1 – Constitution & Volunteer Registration:** Drafted the Eco-Club constitution and enrolled 30 dedicated Standard XI student volunteers across Arts, Science, and Commerce streams.
- **Phase 2 – Activity 1 Setup (Waste Audit):** Procured color-coded segregation bins (Green for wet organic waste, Blue for dry recyclables, Red for e-waste/hazardous), placed them across campus hotspots, and trained canteen workers.
- **Phase 3 – Activity 2 Setup (Herbal Garden):** Cleaned construction debris from the designated 500-sq-ft site, tilled the soil with cow dung manure, installed low-cost drip pipes, and procured native saplings from a government nursery.
- **Phase 4 – Weekly Monitoring & Record Keeping:** Organized weekly Saturday morning maintenance sessions to weigh segregated dry plastics and water herbal plants.
- **Phase 5 – Data Consolidation & Synthesis:** Compiled quantitative project logs and mapped results against Chapter 7 EVS curricular goals.

All activities were monitored and certified by the supervising EVS faculty coordinator.

5. Observations (निरीक्षणे)

The 90-day operational log of the Eco-Club recorded measurable improvements in campus waste diversion and biodiversity:

5.1 Quantitative Project Execution Log

Club Project Phase	Key Operations Conducted	Quantitative Milestone Documented
Single-Use Plastic Audit	Weekly audits of canteen and classrooms; installed 3-stream color-coded dustbins across 6 campus corridors.	Collected and diverted 48 kg of dry multi-layered plastic wrappers; dispatched to authorized municipal recyclers.
Campus Paper Recycling	Established classroom collection boxes for one-sided printed exam sheets, notebooks, and discarded project drafts.	Collected 94 kg of waste paper; handed over to local paper recycling mill for pulp recovery.
Native Herbal Garden	Planted 25 indigenous species: Tulsi, Adulsa, Neem, Curry Leaf, Aloe Vera, Lemongrass, Hibiscus, and Ashwagandha.	Achieved 96% sapling survival rate (24 of 25 plants thriving after 90 days); documented 8 butterfly species visiting flowers.
Campus Composting Pit	Constructed a 4x3 ft aerobic compost pit utilizing fallen dry leaves and shredded canteen vegetable peelings.	Produced ~42 kg of rich, odor-free organic compost utilized directly to fertilize campus garden beds.

5.2 Avian & Insect Biodiversity Observed in Garden

- *Butterflies Recorded:* Common Mormon, Plain Tiger, Lime Butterfly, Grass Yellow.
- *Avian Visitors:* Purple Sunbird, Red-vented Bulbul, Oriental Magpie-Robin.

6. Analysis of data (माहितीचे विश्लेषण)

Analytical evaluation of the Eco-Club's operational data demonstrates the high efficacy of micro-scale campus interventions:

Core Analytical Inferences :-

- **Infrastructure Dictates Waste Behavior:** Pre-audit surveys showed that 82% of campus littering occurred because bins were unsegregated and poorly located. Installing clearly labeled color-coded bins with bilingual signage reduced mixed littering by 70% within four weeks.
- **The Untapped Value of Organic Waste:** College canteens generate substantial quantities of vegetable trimmings and tea leaves that decompose into methane in open dumps. Diverting this material into a simple aerobic compost pit eliminated waste transport costs while generating zero-cost organic fertilizer.
- **Micro-Habitat Creation in Urban Deserts:** Planting nectar-rich native species (Hibiscus, Lantana native, Tulsi) rather than sterile lawn grass attracted eight butterfly species within two months, illustrating how small patches act as vital biodiversity stepping-stones.
- **Peer-Led Environmental Consciousness:** Fellow students responded far more positively to gentle reminders from student Eco-Club volunteers than to administrative disciplinary warnings, proving the power of peer education.

The findings confirm that small-scale student actions create immediate, tangible environmental benefits.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Starting an institutional Eco-Club proved to be an immensely successful action-research project. The junior college campus is noticeably cleaner, boasts an educational medicinal garden, recycles its dry plastics and paper, and produces its own organic compost through decentralized volunteer action.

Action Checklist for Institutional Eco-Clubs :-

- ❑ Formalize an elected student Eco-Club in every junior college with representation from all academic streams.
- ❑ Ban single-use plastic water bottles, disposable cups, and plastic straw usage across college canteens.
- ❑ Establish classroom paper recovery boxes to ensure 100% recycling of term papers and notebook waste.
- ❑ Plant indigenous herbal and flowering plants to transform vacant campus corners into educational nature labs.
- ❑ Integrate Eco-Club field service hours with internal Standard XI EVS project assessment marks.

8. Bibliography / References (संदर्भ सूची)

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